



AI Powered Placement Prediction System

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INTRODUCTION

The Placement Prediction System is a Machine Learning based web application developed to predict whether a student is likely to get placed based on academic performance, communication skills, projects, internships, IQ score, and extracurricular activities.

The system helps students understand their placement readiness and identify areas for improvement.





PROBLEM STATEMENT

01

Students are often unsure about their placement opportunities.

02

Students need a system that can predict their placement chances.

03

Placement depends on factors like CGPA, communication skills, internships, and projects.

04

It is difficult to manually analyze all these factors.



OBJECTIVES

- To predict whether a student will get placed or not.
- To analyze key factors affecting placement such as CGPA, skills, and internships.
- To build a machine learning model for accurate prediction.
- To help students understand their placement readiness.
- To provide basic insights for improving employability.





SYSTEM ARCHITECTURE



01 Input & Preprocessing

Student data (CGPA, skills, internships, projects) is collected and preprocessed by cleaning, encoding, and feature selection.

02 Model Processing

The processed data is passed to a trained Random Forest machine learning model for analysis and prediction.

03 Result Interpretation

The system predicts placement status (Placed / Not Placed) and displays results through the user interface (Streamlit dashboard).



DATASET DESCRIPTION

- Source: Kaggle Placement Dataset
- Contains student academic and skill-related data
- Includes features like CGPA, internships, projects, and communication skills
- Used to train and test the machine learning model
- Helps in predicting placement outcomes accurately





MODEL TRAINING

- Random Forest Classifier is used as the algorithm.
- The dataset is split into training and testing sets.
- The model is trained using student performance data.
- Accuracy score is used to evaluate model performance.
- The best-performing model is selected for final prediction.





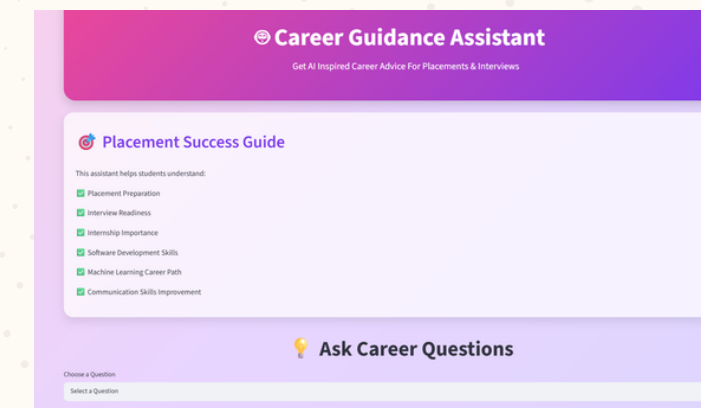
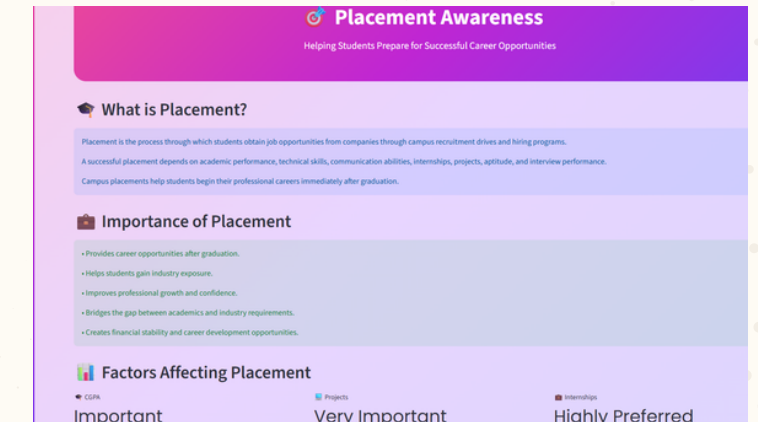
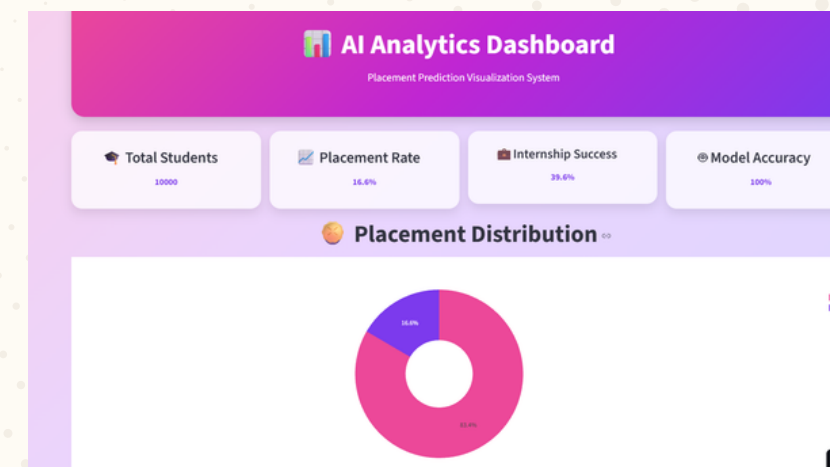
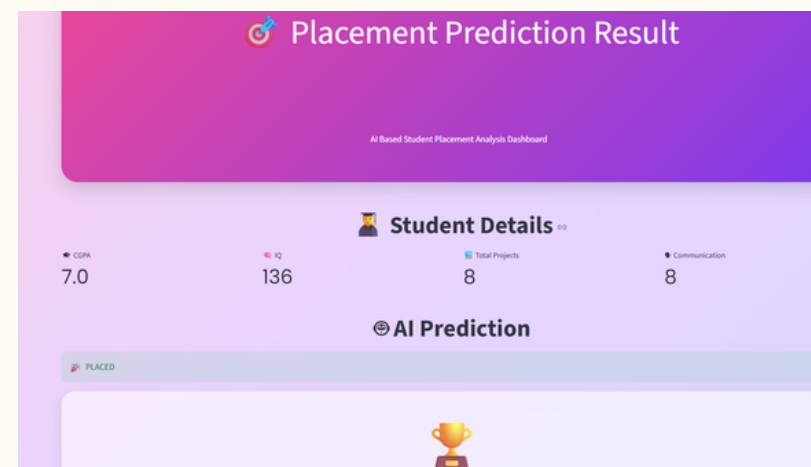
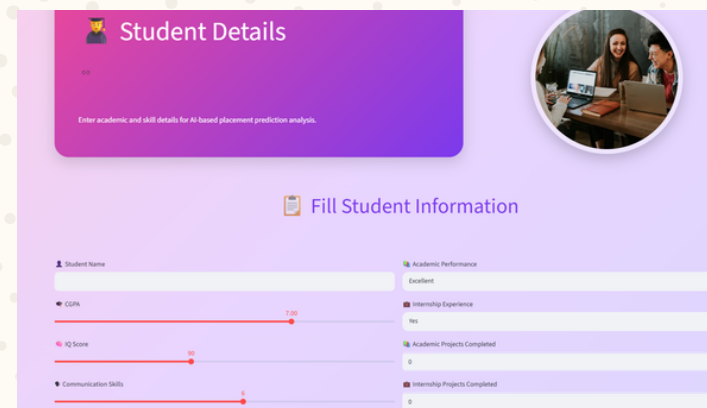
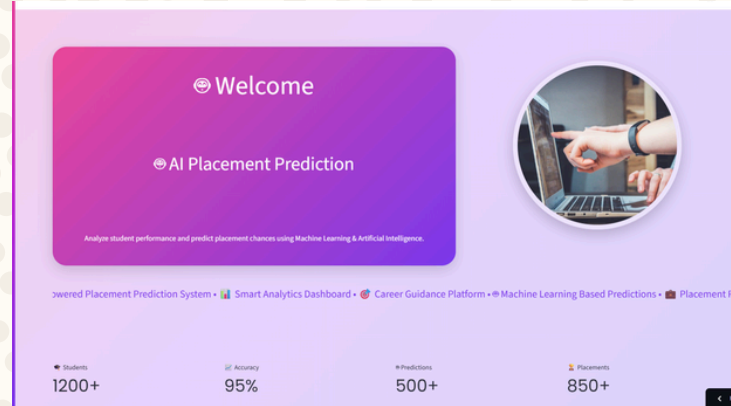
APPLICATION MODULES

- **Home Module** – Provides an overview of the Placement Prediction System and its features.
- **Student Input Module** – Collects student details such as CGPA, skills, internships, and projects.
- **Prediction Module** – Uses the Random Forest model to predict placement status.
- **Visualization Module** – Displays charts and graphical insights for better data analysis.
- **Career Guidance Module** – Provides recommendations to help students improve their employability.
- **Placement Awareness Module** – Highlights key factors that influence placement opportunities and readiness.



MODULE SHOWCASE

The system has different modules that work together for prediction. The Input Module collects student details, the Processing Module prepares the data, and the Prediction Module uses the Random Forest model. The Output Module shows whether the student is placed or not, and the Visualization Module displays results in graphical form.





Live Application

The Placement Prediction System has been successfully deployed as a web application using Streamlit. Users can access the system online, enter student details, and receive real-time placement predictions through an interactive and user-friendly interface.

Project Link:

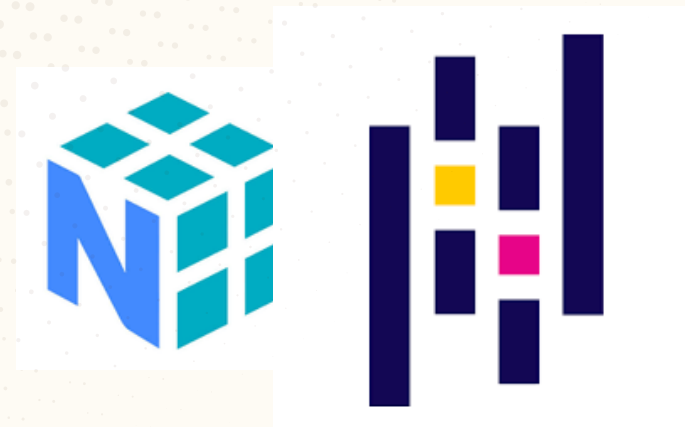
<https://machinelearningproject-internship.streamlit.app/>



TECHNOLOGIES USED



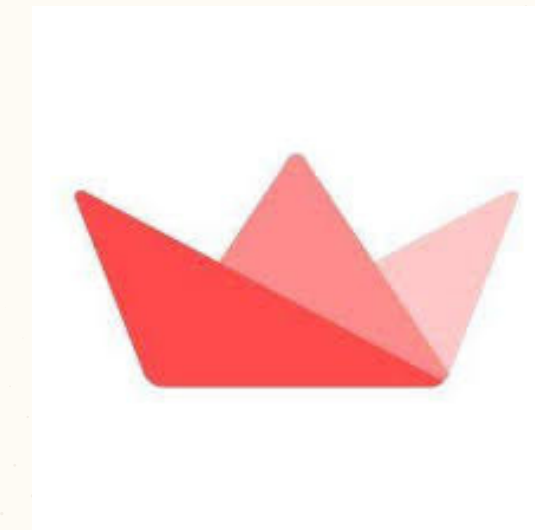
Python
(Programming Language)



Pandas & NumPy
(Data Handling)



Scikit-learn
(Machine Learning Model)



Streamlit
(Web Application)



Matplotlib & Plotly
(Data Visualization)



RESULTS & CONCLUSION

The Placement Prediction System successfully predicts whether a student is likely to be placed based on academic and skill-related factors. The Random Forest model achieved high prediction accuracy and provided reliable results. The system offers valuable insights that help students understand their placement readiness and identify areas for improvement. By using Machine Learning, the prediction process becomes faster, more efficient, and more accurate.





THANK YOU

Thank you for your concern

